



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to Part I (General Handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

Course No. : CHE F314
Course Title : Process Design Principles I
Instructor-in-Charge : SURESH GUPTA
Instructor (Tutorials) : Jay Pandey, Aditya Singh

Course Description

Process invention using heuristics and analysis (The Design process, Process creation and heuristics for process synthesis, Role of process simulators like Aspen Plus in process creation), Strategic designs in process synthesis and analysis, Detailed process synthesis using algorithmic methods with emphasis on reactor networks, separation trains, batch processes, heat and mass integration, energy integration analysis; pinch technology.

Scope & Objective

This course introduces the student to process engineering and provides an insight into various aspects of process design. This course mainly deals with the process synthesis part of process design. It covers in detail the preliminary process synthesis, developing base case design, and using heuristics to choose the best from various process alternatives.

Learning Outcomes

- Strategy of process engineering and an insight into various aspects of process design.
- Use of process simulators for process creation.
- Preliminary process synthesis with the use of rules of thumb or heuristics.
- Sequencing of separation systems for the separation of multicomponent mixtures.
- Heat exchanger network (energy integration analysis) design using Pinch Technology.

Text Books

- T1 Warren D. Seider, J. D. Seader, and Daniel R. Lewin, "Product & Process Design Principles: Synthesis, Analysis, and Evaluation", Wiley-India Edition, India, 2nd Edition (2004).
- T2 James M. Douglas, "Conceptual Design of Chemical Processes", McGraw Hill, New York, International Edition (1988).

Reference Books:

- R1 Richard Turton, Richard C. Bailie, Wallace B. Whiting, Joseph A. Shaeiwitz, and Debangsu Bhattacharyya, "Analysis, Synthesis, and Design of Chemical Processes", International Edition, Pearson Education International, New Jersey (2012).
- R2 Robin Smith, "Chemical Process Design", International Editions, McGraw Hill, Singapore (2000).
- R3 Dale F. Rudd, and Charles C. Watson, "Strategy of Process Engineering", John Wiley & Sons, New York (1968).



Course Plan

Module	Lecture session/Tutorial session	References	Learning outcomes
Introduction to process synthesis and analysis	L1. Introduction to Process Design Principles: Process Synthesis and Analysis.	Ch. 1 of T1 and T2	• Apply the creative aspects and generate ideas for producing new materials & upgrading the existing materials • Develop hierarchical approach to conceptual design and demonstrate using a case study • Generate all the five levels of hierarchy of process flow sheets from the detailed flow sheet
	L2. Creative aspects, Hierarchical approach to conceptual design, HDA Process.	Ch. 3 of T1 and Ch. 1 of T2	
	L3. Preliminary Process Synthesis with examples.		
	T1. Practice problem on Hierarchical approach to conceptual design		
Economic Decision Making	L4. Engineering Economics for Conceptual Design, important cost terms	Ch. 2 of T2	• Collect the cost information required and to calculate capital and operating costs using different methods as well as using simplified cost models depending upon the data availability • List and compare the possible alternatives for a given case study with the help of important aspects associated with the economic decision making • Apply the design aspects associated with various unit operations and to discuss the effect of various design variables on equipment design considerations • Carryout the sensitivity analysis with respect to design variables & and process parameters and to cite the limitations of a rules of thumb (heuristics) for various possible cases
	L5-9. Design of a solvent recovery system, Problem definition, Design of a gas absorber, Flow sheet, material and energy balances, and stream costs, Equipment design considerations, Rules of thumb.	Ch. 3 of T2	
Process Simulation	L10-11. Concepts of process simulation, Steady and unsteady state simulation. Introduction to Aspen Plus.	Ch. 4 of T1 and Class notes	• Initial concept of process simulator in obtaining data and carrying out material and energy balance. • Development of process flow sheet in ASPEN HYSIS software.
Input Information and Batch vs. Continuous	L12-14. Input information, Level-1 decision (Batch vs. continuous), Systematic procedure for the design of batch processes. Economic Potential (EP-1).	Ch. 4 of T2	• List the required input information for a given problem and to demonstrate the importance of input information at the start of conceptual design • To apply the heuristics and to propose the guidelines for selection of batch processes vs. continuous processes
Input-	L15-18. Level-2 Decisions	Ch. 5 of T2	• Apply the decisions required to be made for



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Output Structure of the Flow sheet	(Input-output structure), Design variables, Selectivity and reaction stoichiometry, Overall material balances, Stream		different aspects at level-2 (Input-Output structure of the flow sheet) of economic decision making Identify the possible design variables and to list the information required to develop the
	costs, Process alternatives, Economic potential (EP-2).		material balances at level-2 (input-output structure) Estimate the economic potential at level-(EP2) and to find alternatives at this level by relaxing some of the assumption made
Recycle Structure of the Flow sheet	L19-23. Level-3 Decisions (Recycle structure), Recycle material balances, Reactor heat effects, Equilibrium limitations, Compressor design and costs, Reactor design, Economic potential (EP-3).	Ch. 6 of T1 and Ch. 6 of T2	- Estimate the number of reactor systems and the number of recycle streams required for a given process at level-3 (recycle structure) - Develop the recycle material balances and to compute the various flows associated with a recycle structure of process flow sheet - Estimate the economic potential at level-3 (EP3) and to show the effect of design variables on EP3
Separation System	L24-26. General structure, Vapor recovery system, Liquid separation system, Azeotropic systems, Rigorous material balances, Economic potential (EP-4).	Ch. 7 of T1 and Ch. 7 of T2	- Generate the possible alternatives for the general structure of flow sheet and to examine the phases of the reactor effluent stream - Discuss the decisions associated with vapor recovery systems and to find the sequence of separation systems - Develop rigorous material balances (linear material balancing) and to estimate the economic potential at level-4 (EP4) for the general structure of a flow sheet
Energy Integration Analysis	L27-37. Heat-Exchanger Network Synthesis (HENS), First law analysis, Cascade diagrams, Temperature-Enthalpy Diagrams, Grand composite curve, Multiple utilities, Area estimates, Design of MER Networks, Loops and paths, Final design, Stream splitting, Complete design algorithm, Heat and power integration.	Ch. 10 of T1 and Ch. 8 of T2	- Estimate the net heat available in the given process plant using I & II law of thermodynamics - Identify the appropriate pinch and hot & cold utility requirements from the cascade diagram - Draw the temperature-Enthalpy diagram and estimate the total area requirement, <i>a priori</i>, network proposal - Generate design alternatives in terms of feasible matches in the network and propose the MER (maximum energy recovery or minimum energy requirement) network - Propose the final Heat Exchanger Network after breaking all the loops and restore ΔT_{min} as and when there is a violation
Mass Integration	L38-40. Introduction, Minimum mass separation agent, Mass exchanger networks.	Ch. 11 of T1	- Able to compute the minimum usage of external mass-separating agent (MSA) - Able to design mass exchanger network which meets minimum operating cost target.



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Evaluation Scheme

EC No.	Evaluation component (EC)	Duration (Minutes)	Marks (300)	Date and time	Nature of component
1	Mid Semester Test	90	90	10/10 FN1	Closed Book
2	Class Quizzes (Surprise) #	-	30	-	Closed/Open Book
3	ASPEN Assignment + related quizzes*		20 + 40	To be announced in Nalanda Portal	Open Book
4	Comprehensive Examination	120	120	15/12 FN	Closed Book

#Total **eight class quizzes** will be conducted. Out of these **eight quizzes**, the performance of the **best six** will be considered for final evaluation.

*Conceptual design oriented problems/case studies will be given as ASPEN Assignment. Also, there will be continuous evaluation in the tutorials related to ASPEN-HYSYS. **There will be three quizzes based ASPEN-HYSYS and out of these best two** will be considered for final evaluation. **Exact dates for ASPEN-HYSYS quizzes will be announced in the class in priori on the Nalanda Portal.**

Chamber Consultation Hour: To be announced in the class.

Make-up policy: No make-up will be granted for quizzes (Class + ASPEN-HYSYS). Make-up for Mid Semester Test and Comprehensive Examination will be granted only for genuine reason(s) for not appearing in the regular test. Proper proofs must be submitted along with Make-up application. **Prior permission of IC is compulsory.**

Notices: All notices concerning this course will be made available online on the Nalanda Portal.

Instructor-in-charge
CHE F314